Candidate Seat No: __________

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Bachelor of Computer Applications (BCA)
Semester-II University Examination May 2018
CA 108 – Mathematical and Statistical Foundations

Date: 21/05/2018, Monday **Time: 10:00 a.m. to 01:00 p.m.** **Maximum Marks: 70**

Instructions:
1. **Attempt the two sections in separate answer books.**
2. **Figure to the right indicate marks.**
3. **Make suitable assumptions wherever necessary and state them.**

SECTION I

Q.1 Do as directed.

1. A set which having a single element is known as equal set. **(True/False)** **[01]**
2. A matrix with the same number of rows and columns is said to be _____ matrix. **[01]**
3. $\int 5x^3 \; dx = \frac{5x^4}{4} + C$. **(True/False)** **[01]**
4. If U={1,2,3,4,5,6} and B={1,2,4,6}, then B'=_______________. **[01]**
5. If y = x^5 then $\frac{dy}{dx}$ = _______. **[01]**
6. f(x) = 3x + 3, D*f* = {5,2}. Find R*f*. **[02]**
7. If A={1,2} and B={n}, then Find A x B . **[02]**
8. Find determinant value of A= $\begin{vmatrix} 1 & 0 & 0 \\ 1 & 2 & 1 \\ 1 & 0 & 2 \end{vmatrix}$ **[02]**

Q.2 Answer the following. (Attempt any three) **[12]**

[A] U = {a,b,c,d,e,f,g,h} A = { a,c,e,f,h} & B = { a,b,c,e,g} . Find A' ∪ B and A ∩ B' .

[B] Find derivative of y = $(3x^2 + 2)(2x + 3)$

[C] Find B + A and A – B , if A = $\begin{bmatrix} 2 & 6 & 0 \\ 2 & 4 & 6 \\ 2 & 6 & 4 \end{bmatrix}$ & B = $\begin{bmatrix} 1 & 3 & 0 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \end{bmatrix}$

[D] Define the following terms with example:
- Square Matrix
- Singleton set

[E] Evaluate $\lim_{n \to \infty} \frac{15n^2 - 14n + 1}{5n^2 + 5n - 2}$

Q.3 Answer the following. (Attempt any two) **[12]**

[A] Find A' – B and A Δ B', if U={1,2,3,4….10} A={1,3,5,6,7} & B= {1,2,5,6,8}

[B] Find AB , if A = $\begin{bmatrix} 5 & 3 & 0 \\ 1 & 0 & 3 \\ 1 & 3 & 4 \end{bmatrix}$ & B = $\begin{bmatrix} 2 & 3 & 0 \\ 5 & 1 & 3 \\ 0 & 3 & 2 \end{bmatrix}$

[C] If f(x) = $\frac{x^2 + 2x}{x + 2}$, find f(1) +f(3) + f(2)

[D] Evaluate $\int_1^4 (5x^2 - 8x + 5)dx$

SECTION II

Q.4 Do as directed.

1. What is the probability of getting a red cards from pack of playing cards? **[01]**

2. Range is a measure of central tendency. (True/False) **[01]**

3. _______________ approach helps decision makers in making rational and effective decisions. **[01]**

4. The median is most commonly used to find the average value within the distribution. (True/False) **[01]**

5. $^5P_2$ = __________. **[01]**

6. Find Range and Mode for following marks : 2, 3, 7, 2, 3, 6, 2, 7, 5. **[02]**

7. Find out Mean and Median for observations : 1, 2, 3, 4, 5, 4, 3, 2, 1. **[04]**

Q.5 Answer the following. (Attempt any two) **[12]**

[A] Formulate the LPP (Linear Programing Problem)

- Profit maximization problem that our manufacturer faces. The company uses different materials and labor to produce Table and Chair. Recall that unit profit for Table is $6, and unit profit for Chair is $8. There are 300 feet (f) of material available, and 110 hours of labor available. It takes 30 f and 5 hours to make Table, and 20 f and 10 hours to make Chair.

Resource	Table (X1)	Chair (X2)	Available
Material	30	20	300
Labor	5	10	110
Profit	6	8	

[B] Calculate Mode and Median.

Marks (x)	0-20	20-40	40-60	60-80	80-100
No. of students	5	8	15	16	6

[C] How many different arrangements can be made using all the letters of the word "TUESDAY" for following cases ? (any two)

1. Start with T and end with Y.
2. UES Must be together.
3. Begin with vowels.
4. Consonants must come together.

[D] Find Standard deviation & Variance for observations : 1, 2, 3, 4, 5, 4, 3, 2, 1, 5.

Q.6 Answer the following. (Attempt any two) **[12]**

[A] Write a short note on components of LPP (Linear Programming Problem).

[B] List types of events in probability and explain each one in detail with examples.

[C] Calculate Mean deviation.

Marks (x)	0-10	10-20	20-30	30-40	40-50
No. of students	15	10	25	30	20

[D] A box contains 2 books of C programming, 3 books of C++ & 4 books of Java books, Find no. of ways in which 3 books can be select, if **(Any Two)**

1. Do not select Java books.
2. Select only Java books.
3. An equal number of Book get selected.
4. Select 2 C books and 1 C++ book.